

Information Search Retrieval Through BM25 and TF-IDF Ranking Models

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I. INTRODUCTION

This assignment implements an information retrieval system by utilizing the BM25 and TF-IDF ranking models. The pages pulled from the built web crawler in the previous assignment were the files collected and ranked by these models based on user queries. This was done through an inverted index from the corpus and two distinct classes for indexing and searching.

II. IMPLEMENTATION

Implementation was performed through the Indexer and SearchAgent classes.

A. Indexer Class

The Indexer class is composed of a constructor and methods to create the index, load the corpus, preprocess text, and save and load indexes. Creating the index involves looping through corpus documents, preprocessing their text, adding their terms to the vocabulary, and adding them to the inverted index. Loading the corpus simply involves looping through the files and making the appropriate document objects to add to the corpus. Text preprocessing involves tokenization, stopword removal, and eliminating extra spaces and characters. Saving and loading the indexes either creates a data dictionary or loads the existing one.

B. SearchAgent Class

The SearchAgent class consists of a constructor and methods for BM25 query, TF-IDF query, and displaying results. The query functions rank documents by retrieving preprocessed text, creating dictionaries, then looping through each term in the user query to find its location in the inverted index, its document frequencies, and using these for their respective ranking formulas. Results are displayed by looping through these rankings and printing the document characteristics of each.

III. RANKING METHODS

A. TF-IDF

The TF-IDF model ranks documents based on the product of term frequency and inverse document frequency. Document scores for a query are computed as the sum of TF-IDF scores for all query terms appearing in that document.

```
Enter query: computer science department
bm25 or tfidf?bm25
DocID: 198
Score: 17.519125379825212
URL: https://nlp.stanford.edu/IR-book/information-retrieval-more-books.html
filename: doc_277.json

DocID: 236
Score: 14.74949077711338
URL: http://nlp.stanford.edu/IR-book/html/htmledition/document-delineation-and-character-sequence-decoding-1.html
filename: doc_41.json

DocID: 226
Score: 14.344189023054561
URL: http://nlp.stanford.edu/IR-book/html/htmledition/prerequisites-1.html
filename: doc_32.json

DocID: 79
Score: 13.833642399206
URL: http://nlp.stanford.edu/IR-book/html/htmledition/text-classification-and-naive-bayes-1.html
filename: doc_17.json

DocID: 0
Score: 11.999609941316553
URL: https://nlp.stanford.edu/IR-book/information-retrieval-book.html
filename: doc_0.json
```

Fig. 1. BM25 Query

```
Enter query: computer science department
bm25 or tfidf?tfidf
DocID: 79
Score: 178.29894404833484
URL: http://nlp.stanford.edu/IR-book/html/htmledition/text-classification-and-naive-bayes-1.html
filename: doc_17.json

DocID: 195
Score: 53.303401670795544
URL: http://nlp.stanford.edu/IR-book/html/htmledition/references-and-further-reading-21.html
filename: doc_274.json

DocID: 234
Score: 47.032413238937245
URL: http://nlp.stanford.edu/IR-book/html/htmledition/the-extended-boolean-model-versus-ranked-retrieval-1.html
filename: doc_4.json

DocID: 203
Score: 45.98492383457062
URL: http://nlp.stanford.edu/IR-book/html/htmledition/a-first-take-at-building-an-inverted-index-1.html#sec:indexfirsttake
filename: doc_281.json

DocID: 231
Score: 45.98492383457062
URL: http://nlp.stanford.edu/IR-book/html/htmledition/a-first-take-at-building-an-inverted-index-1.html
filename: doc_37.json
```

Fig. 2. TF-IDF Query

B. BM25

BM25 is a ranking function that considers term frequency, inverse document frequency, and document lengths. It is better than TF-IDF at reducing the bias toward longer documents and putting more focus on term frequency based on document length.

C. Query Test

D. Result Comparison

From the query test, BM25 clearly handles longer documents better and prioritizes term frequency by the length of

the document, leading it to yield better results. TF-IDF, on the other hand, is much simpler and does not have good resistance to long documents, neglecting good term frequency handling as well. BM25 is therefore more balanced. TF-IDF assigns higher scores to longer documents and does not take into account this length when locating term frequencies. BM25 solves this through its normalization process.

IV. CHALLENGES

Some challenges included implementing the BM25 formula correctly as well as ensuring proper and consistent preprocessing for documents and queries. The ranking formula was tricky because of its complexity to translate into code. Meanwhile, preprocessing was essential to ensure matches could be found in documents from queries. Without consistent preprocessing, this would not have been possible.

V. CONCLUSION

This assignment demonstrates the creation of a basic information retrieval system using an inverted index and proper vocabulary, querying, and preprocessing techniques. The BM25 and TF-IDF ranking models were put into use and subsequently evaluated, with BM25 yielding the better search results due to its balanced nature. A few challenges were present in the assignment, but they were able to be managed to properly complete the tasks necessary.